HAT-P-12b

R-1469

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-12_220606` · created 2026-08-01T11:37:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459736.8136 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.052
Transit depth (Rp/Rs)^2	3.49 +/- 1.93 [%]
Orbital Inclination (inc)	86.38 +/- 1.56
Airmass coefficient 1 (a1)	0.922 +/- 0.038
Airmass coefficient 2 (a2)	0.057 +/- 0.029
Scatter in the residuals of the lightcurve fit is	4.16 %
Best Comparison Star	#2 - [427, 251]
Optimal Aperture	2.3
Optimal Annulus	7.52
Transit Duration (day)	0.072 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.187 ± 0.052 vs 0.1406 (deviation 0.78σ)
Duration fit vs published	0.072 d vs 0.0975 d (deviation 0.77σ)
Mid-time O–C	9.5 min
Depth SNR	3.1
Residual scatter	4.16 %

Light curve

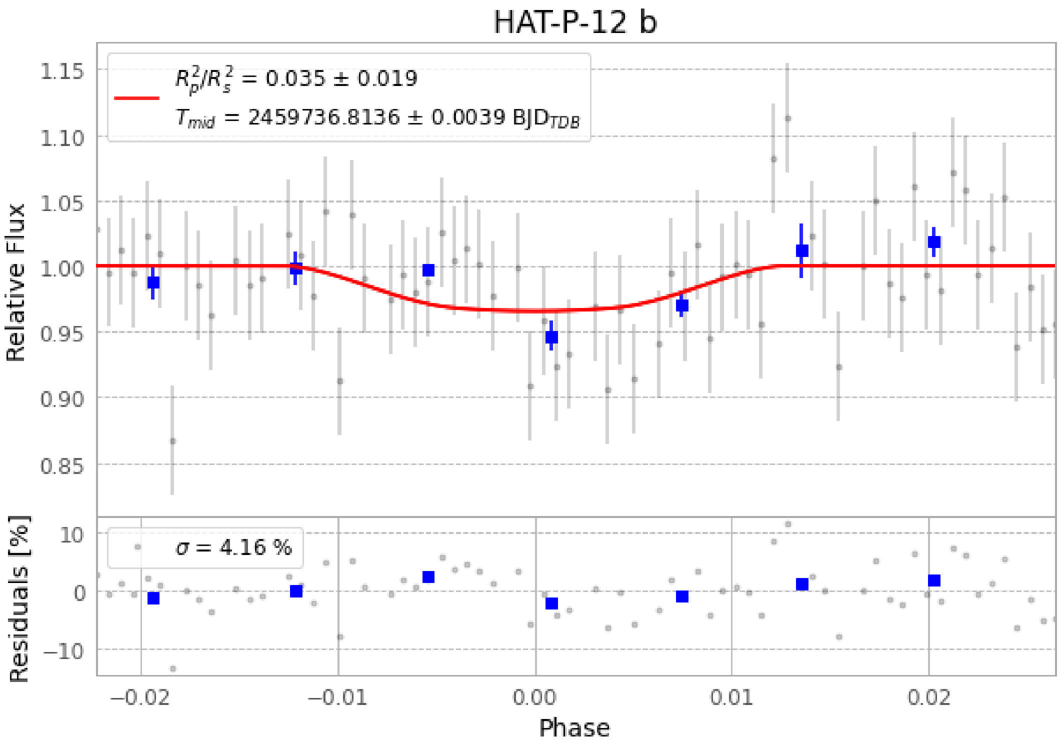

Figure 1 — FinalLightCurve_HAT-P-12 b_2022-06-05.png (SHA-256 58e208d962249f6a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 209], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2cd848cf37a7659c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

76 FITS frames (50 MB), `HATP-12220606054513.FITS` → `HATP-12220606093012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-220606.log (17ae80278396…), FinalLightCurve_HAT-P-12 b_2022-06-05.png (58e208d96224…), FinalLightCurve_HAT-P-12 b_2022-06-05.csv (d58f6095a32f…)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)